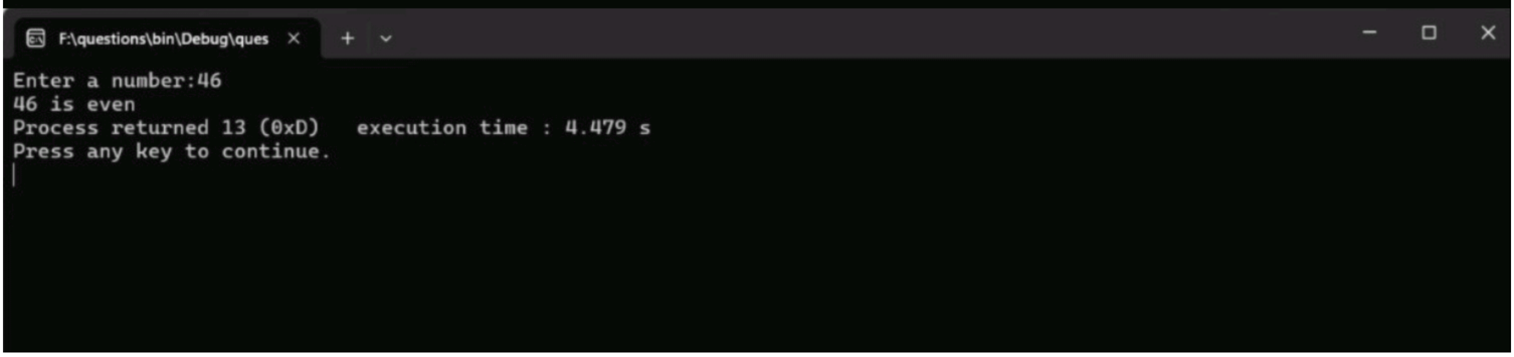
```c
#include<stdio.h>
void main()
{
    printf("Hello World");
getch();
}
```

```c
#include<stdio.h>
void main()
{
    int num;
    printf("Enter a number:");
    scanf("%d",&num);
    if(num%2==0)
    {
        printf("%d is even",num);
    }
    else
    {
        printf("%d is odd",num);
    }
getch();
}
```


F:\questions\bin\Debug\ques
Hello World


F:\questions\bin\Debug\ques
Enter a number:46
46 is even
Process returned 13 (0xD)   execution time : 4.479 s
Press any key to continue.

```c
#include<stdio.h>
void main()
{
    int a,b,c;
    printf("Enter any three numbers:");
    scanf("%d %d %d",&a,&b,&c);
    if(a>=b)
    {
        if(a>=c)
        printf("%d is the largest number",a);
        else
        printf("%d is the largest number",c);
    }
    else
    {
        if(b>=c)
        printf("%d is the largest number",b);
        else
        printf("%d is the largest",c);

    }
getch();
}
```

```c
#include<stdio.h>
void main()
{
    int year;
    printf("Enter a year:");
    scanf("%d",&year);
    if(year%4==0 && year%100!=0 || year%400==0)
    {
        printf("%d is a Leap Year",year);
    }
    else
    {
        printf("%d is Not a Leap Year",year);
    }
getch();
}
```

```
Enter any three numbers:32
56
7
56 is the largest number
Process returned 13 (0xD)   execution time : 10.328 s
Press any key to continue.
```

```
Enter a year:2013
2013 is Not a Leap Year
Process returned 13 (0xD)   execution time : 7.735 s
Press any key to continue.
```

```
Enter a year:1980
1980 is a Leap Year
Process returned 13 (0xD)   execution time : 15.809 s
Press any key to continue.
```

```c
#include<stdio.h>
void main()
{
    int num;
start:
    printf("Enter a positive number:");
    scanf("%d",&num);
    if(num<=0)
    {
        printf("Enter postive number \n");
        goto start;
    }
    printf("%d is a positive number",num);
getch();
}
```

```c
#include<stdio.h>
void main()
{
    int choice;
    float num1,num2,result;
    printf("Enter your choice(1-add,2-subtract,3-Multiply,4-Divide)");
    scanf("%d",&choice);
    printf("Enter two numbers");
    scanf("%f %f",&num1,&num2);
    switch(choice)
    {
        case 1 : result = num1+num2;
                 printf("Addition is %f",result);
                 break;
        case 2 : result = num1-num2;
                 printf("Subtraction is %f",result);
                 break;
        case 3 : result = num1*num2;
                 printf("Multiplication is %f",result);
                 break;
        case 4 :
                 if(num2==0)
                 {
                     printf("Division not allowed");
                 }
                 else
                 {
                 result = num1/num2;
                 printf("Division is %f",result);
                 }
                 break;
        default:
                 printf("Invalid input");
    }
    getch();
}
```

```
Enter a positive number:0
Enter postive number
Enter a positive number:-2
Enter postive number
Enter a positive number:57
57 is a positive number
Process returned 13 (0xD)   execution time : 16.654 s
Press any key to continue.
```

```
Enter your choice(1-add,2-subtract,3-Multiply,4-Divide)3
Enter two numbers17
25
Multiplication is 425.000000
Process returned 13 (0xD)   execution time : 22.008 s
Press any key to continue.
```

```c
#include<stdio.h>
void main()
{
     int num,i,k;
     printf("Enter a number:");
     scanf("%d",&num);
     for(i=1;i<=10;i++)
     {
         k=num*i;
     printf("%d*%d=%d",num,i,k);
     printf("\n");
     }
getch();
}
```

```c
#include<stdio.h>
void main()
{
    int num,k=1,i=1;
    printf("Enter a positive number:");
    scanf("%d",&num);
    while(i<=num)
    {
        k=k*i;
        i++;
    }
    printf("Factorial of %d is %d",num,k);
getch();
}
```

```
Enter a number:18
18*1=18
18*2=36
18*3=54
18*4=72
18*5=90
18*6=108
18*7=126
18*8=144
18*9=162
18*10=180

Process returned 13 (0xD)   execution time : 6.925 s
Press any key to continue.
```

```
Enter a positive number:6
Factorial of 6 is 720
Process returned 13 (0xD)   execution time : 3.401 s
Press any key to continue.
```

```c
#include<stdio.h>
void main()
{
 int n,i,a=0,b=1,c;
 printf("Enter number of terms required:");
 scanf("%d",&n);
 printf("Fibonacci Series: ");
 for(i=1;i<=n;i++)
 {
     printf("%d ",a);
     c=a+b;
     a=b;
     b=c;
 }
getch();
}
```

```c
#include<stdio.h>
void main()
{
 int r,q,n,s=0;
 printf("Enter a number:");
 scanf("%d",&n);
 do
 {
     r=n%10;
     q=n/10;
     n=q;
     s=r+s;
 }
 while(n>0);
 printf("Sum of digits is %d",s);
getch();
}
```

```
Enter number of terms required:9
Fibonacci Series: 0 1 1 2 3 5 8 13 21
Process returned 13 (0xD)   execution time : 7.942 s
Press any key to continue.
```

```
Enter a number:2345
Sum of digits is 14
Process returned 13 (0xD)   execution time : 13.851 s
Press any key to continue.
```

```c
#include<stdio.h>
void main()
{
 int r,q,num,d=0,x;
 printf("Enter a number:");
 scanf("%d",&num);
 x=num;
 while(num>0)
 {
      r=num%10;
      q=num/10;
      num=q;
      d=d*10+r;
 }
 printf("Reverse of %d is %d",x,d);
getch();
}
```

```c
#include<stdio.h>
void fact(int k)
{
    int i,n=1;
    for(i=1;i<=k;i++)
    {
        n=n*i;
    }
    printf("Factorial of %d is %d",k,n);
}
int main()
{
    int num;
    printf("Enter a number:");
    scanf("%d",&num);
    fact(num);
    return 0;
}
```

```
Enter a number:4569
Reverse of 4569 is 9654
Process returned 13 (0xD)   execution time : 17.118 s
Press any key to continue.
```

```
Enter a number:7
Factorial of 7 is 5040
Process returned 0 (0x0)   execution time : 3.224 s
Press any key to continue.
```

```c
#include<stdio.h>
void swap(int *x,int *y)
{
    int temp;
    temp=*x;
    *x=*y;
    *y=temp;
}
int main()
{
    int a,b;
    printf("Enter any two numbers:");
    scanf("%d %d",&a,&b);
    printf("\nBefore swapping:a=%d,b=%d",a,b);
    swap(&a,&b);
    printf("\nAfter swapping:a=%d,b=%d",a,b);
    return 0;
}
```

```c
#include<stdio.h>
int main()
{
    void max(int,int);
    int x,y;
    printf("Enter value for x and y:");
    scanf("%d %d",&x,&y);
    max(x,y);
    return 0;
}
void max(int a ,int b)
{
    if(a>b)
        printf("Maximum of two numbers is %d",a);
    else if(b>a)
        printf("Maximum of two numbers is %d",b);
    else
        printf("Numbers are equal");
}
```

```
Enter any two numbers:34
90

Before swapping:a=34,b=90
After swapping:a=90,b=34
Process returned 0 (0x0)   execution time : 5.740 s
Press any key to continue.
```

```
Enter value for x and y:23
67
Maximum of two numbers is 67
Process returned 0 (0x0)   execution time : 6.711 s
Press any key to continue.
```

```
#include<stdio.h>
int main()
{
    void prime();
    prime();
    return 0;
}
void prime()
{
    int i,num,c=0;
    printf("Enter a number:");
    scanf("%d",&num);
    for(i=1;i<=num;i++)
    {
        if(num%i==0)
            c=c+1;
    }
    if(c==2)
    printf("%d is prime",num);
    else
    printf("%d is not prime",num);
}
```

```
#include<stdio.h>
int power(int x,int n)
{
    if(n==0)
        return 1;
    else
        return(x*power(x,n-1));
}
int main()
{
    int x,n,ans;
    printf("Enter base number:");
    scanf("%d",&x);
    printf("Enter exponent:");
    scanf("%d",&n);
    ans=pow(x,n);
    printf("Result is %d",ans);

    return 0;
}
```

```
Enter a number:17
17 is prime
Process returned 0 (0x0)   execution time : 5.154 s
Press any key to continue.
```

```
Enter a number:99
99 is not prime
Process returned 0 (0x0)   execution time : 2.371 s
Press any key to continue.
```

```
Enter base number:5
Enter exponent:4
Result is 625
Process returned 0 (0x0)   execution time : 9.881 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    int arr[5],i,j,big=0;
    for(i=0;i<=4;i++)
    {
        printf("Enter a number:");
        scanf("%d",&arr[i]);
    }
    for(i=0;i<=3;i++)
    {
        for(j=i+1;j<=4;j++)
        {
            if(arr[i]>arr[j])
            {
                big=arr[i];
                arr[i]=arr[j];
                arr[j]=big;
            }
        }
    }
printf("Largest element is %d",big);
return 0;
}
```

```c
#include<stdio.h>
int main()
{
    int arr[5],i,j;
    float sum=0,avg;
    for(i=0;i<=4;i++)
    {
        printf("Enter a number:");
        scanf("%d",&arr[i]);
    }
    for(i=0;i<=4;i++)
    {
        sum=sum+arr[i];
    }
    printf("Sum of elements is %f",sum);
    avg=sum/5;
    printf("\nAverage of elements is %f",avg);
return 0;
}
```

```
Enter a number:23
Enter a number:45
Enter a number:67
Enter a number:89
Enter a number:9
Largest element is 89
Process returned 0 (0x0)   execution time : 6.337 s
Press any key to continue.
```

```
Enter a number:25
Enter a number:46
Enter a number:78
Enter a number:90
Enter a number:10
Sum of elements is 249.000000
Average of elements is 49.799999
Process returned 0 (0x0)   execution time : 8.275 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    int arr[5],i,j,search;
    for(i=0;i<=4;i++)
    {
        printf("Enter a number:");
        scanf("%d",&arr[i]);
    }
    printf("Enter number to be searched:");
    scanf("%d",&search);
    for(i=0;i<=4;i++)
    {
      if(arr[i]==search)
      {
          printf("%d is found at element %d",search,i+1);
          break;
      }
    }
return 0;
}
```

```c
#include<stdio.h>
int main()
{
    int arr[5],i,j,temp=0,n=5;
    for(i=0;i<n;i++)
    {
        printf("Enter a number:");
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n-1;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if(arr[j]>arr[j+1])
            {
                temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    printf("Sorted array:");
    for(i=0;i<n;i++)
    {
        printf("%d ",arr[i]);
    }
return 0;
}
```

```
Enter a number:34
Enter a number:45
Enter a number:6
Enter a number:78
Enter a number:99
Enter number to be searched:78
78 is found at element 4
Process returned 0 (0x0)   execution time : 10.631 s
Press any key to continue.
```

```
Enter a number:24
Enter a number:35
Enter a number:17
Enter a number:2
Enter a number:9
Sorted array:2 9 17 24 35
Process returned 0 (0x0)   execution time : 13.527 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    int arr[7],i,start,end,mid,search;
    for(i=0;i<7;i++)
    {
        printf("Enter a number:");
        scanf("%d",&arr[i]);
    }
    printf("Enter number to be searched:");
    scanf("%d",&search);
    start=0,end=6;
    for(i=0;i<7;i++)
    {
        mid=(start+end)/2;
        if(arr[mid]==search)
        {
            printf("Search successful at element number %d",i+1);
            break;
        }
        if(arr[mid]<search)
        {
            start=mid+1;
        }
        if(arr[mid]>search)
        {
            end=mid-1;
        }
    }
return 0;
}
```

```c
#include<stdio.h>
int main()
{
    int arr1[3][3],arr2[3][3],arr3[3][3],i,j;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("Enter elements in array arr1:");
            scanf("%d",&arr1[i][j]);
        }
    }
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("Enter elements in array arr2:");
            scanf("%d",&arr2[i][j]);
        }
    }
    printf("Sum of two given matrices:\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            arr3[i][j]=arr1[i][j]+arr2[i][j];
            printf("%d ",arr3[i][j]);
        }
        printf("\n");
    }
return 0;
}
```

```
Enter a number:23
Enter a number:3
Enter a number:4
Enter a number:5
Enter a number:6
Enter a number:7
Enter a number:88
Enter number to be searched:3
Search successful at element number 2
Process returned 0 (0x0)   execution time : 13.286 s
Press any key to continue.
```

```
Enter elements in array arr1:12
Enter elements in array arr1:23
Enter elements in array arr1:24
Enter elements in array arr1:35
Enter elements in array arr1:55
Enter elements in array arr1:6
Enter elements in array arr1:7
Enter elements in array arr1:8
Enter elements in array arr1:9
Enter elements in array arr2:10
Enter elements in array arr2:20
Enter elements in array arr2:30
Enter elements in array arr2:40
Enter elements in array arr2:44
Enter elements in array arr2:55
Enter elements in array arr2:6
Enter elements in array arr2:78
Enter elements in array arr2:90
Sum of two given matrices:
22 43 54
75 99 61
13 86 99

Process returned 0 (0x0)   execution time : 19.295 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    char st[100],ch;
    int v=0,c=0,i;
    printf("Enter a string:");
    gets(st);
    for(i=0;st[i]!='\0';i++)
    {
        ch=st[i];
        if(ch>='a' && ch<='z' || ch>='A' && ch<='Z')
        {
            if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'||
               ch=='A'||ch=='E'||ch=='I'||ch=='O'||ch=='U' )
            {
                v=v+1;
            }
            else
            {
                c=c+1;
            }
        }
    }
    printf("Total Vowels:%d",v);
    printf("\nTotal Consonants:%d",c);
  return 0;
}
```

```c
#include<stdio.h>
int main()
{
    char st[25];
    int count=0,i;
    printf("Enter a string:");
    gets(st);
    for(i=0;st[i]!='\0';i++)
    {
        count=count+1;
    }
    printf("Reverse\n");
    for(i=count-1;i>=0;i--)
    {
        printf("%c",st[i]);
    }
  return 0;
}
```

```
Enter a string:Programming in C
Total Vowels:4
Total Consonants:10
Process returned 0 (0x0)   execution time : 37.063 s
Press any key to continue.
```

```
Enter a string:Good will
Reverse
lliw dooG
Process returned 0 (0x0)   execution time : 18.797 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    char st1[25];
    char st2[25];
    int i=0,j=0;
    printf("Enter string1:");
    gets(st1);
    printf("Enter string2:");
    gets(st2);
    while(st1[i]!='\0')
    {
        i=i+1;
    }
    while(st2[j]!='\0')
    {
        st1[i]=st2[j];
        i++;
        j++;
    }
    st1[i]='\0';
    printf("Concatenated String:%s",st1);
    return 0;
}
```

```c
#include<stdio.h>
int main()
{
    char str[25];
    int i=0,j;
    printf("Enter a string:");
    gets(str);
    while(str[i]!='\0')
    {
        i++;
    }
    j=i-1;
    i=0;
    while(i<j)
    {
        if(str[i]!=str[j])
        {
            printf("Not a palindrome");
            return 0;
        }
        i++;
        j--;
    }
    printf("Palindrome");
return 0;
}
```

```
Enter string1:Programm
Enter string2:ing
Concatenated String:Programming
Process returned 0 (0x0)   execution time : 13.146 s
Press any key to continue.
```

```
Enter a string:madam
Palindrome
Process returned 0 (0x0)   execution time : 4.129 s
Press any key to continue.
```

```c
#include<stdio.h>
int main()
{
    struct student
    {
        int rno,total;
        char name[25];
    };
    struct student x;
    printf("Enter Student Name:");
    gets(x.name);
    printf("Enter Student's RollNo.");
    scanf("%d",&x.rno);
    printf("Enter Total Marks obtained in C out of 100:");
    scanf("%d",&x.total);
    printf("Details Entered are:\n");
    printf("\nRollNo.-%d",x.rno);
    printf("\nStudent Name-%s",x.name);
    printf("\nTotal Marks-%d",x.total);
return 0;
}
```

```c
#include<stdio.h>
int gcd(int a ,int b)
{
    if(b==0)
        return a ;
    return gcd(b,(a%b));
}
int main()
{
    int num1,num2;
    printf("Enter two numbers:");
    scanf("%d %d",&num1,&num2);
    printf("GCD=%d",gcd(num1,num2));
    return 0;
}
```

```
Enter Student Name:Prem
Enter Student's RollNo.22
Enter Total Marks obtained in C out of 100:67
Details Entered are:

RollNo.-22
Student Name-Prem
Total Marks-67
Process returned 0 (0x0)   execution time : 11.691 s
Press any key to continue.
```

```
Enter two numbers:36
18
GCD=18
Process returned 0 (0x0)   execution time : 26.901 s
Press any key to continue.
```